/ - division [Quotient]:

5 / 2 = 2 [int / int = int]

5.0/2=2.500000

5/2.0=2.500000

5.0/2.0=2.500000

Note: In C & C++ both operands are int then answer also int. if anyone or both are float the answer is float.

TC

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Line 11 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
float  a=3/2;
clrscr();
printf("%d\n",5/2);
printf("%f\n",5.0/2);
printf("%f\n",5/2.0);
printf("%f\n",5.0/2.0);
printf("a=%f",a);
}
```

2:27 PM 17-Aug-24

TC

```
2
2.500000
2.500000
2.500000
a=1.000000_
```

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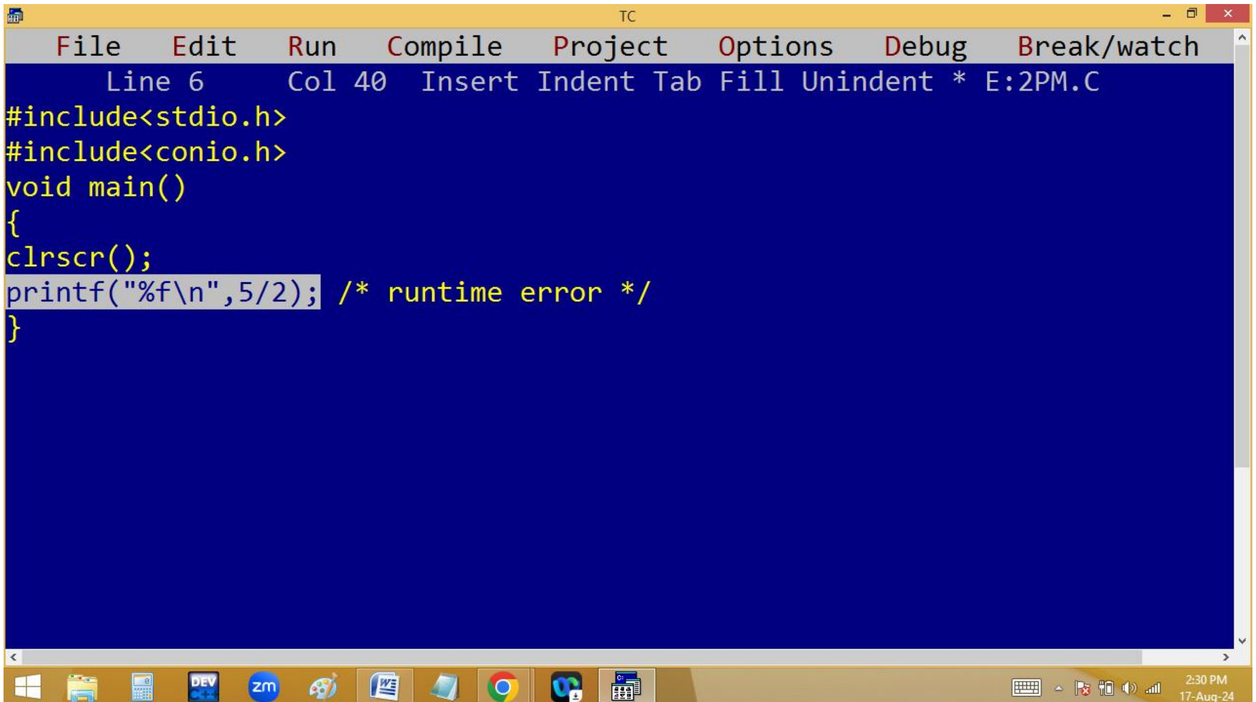
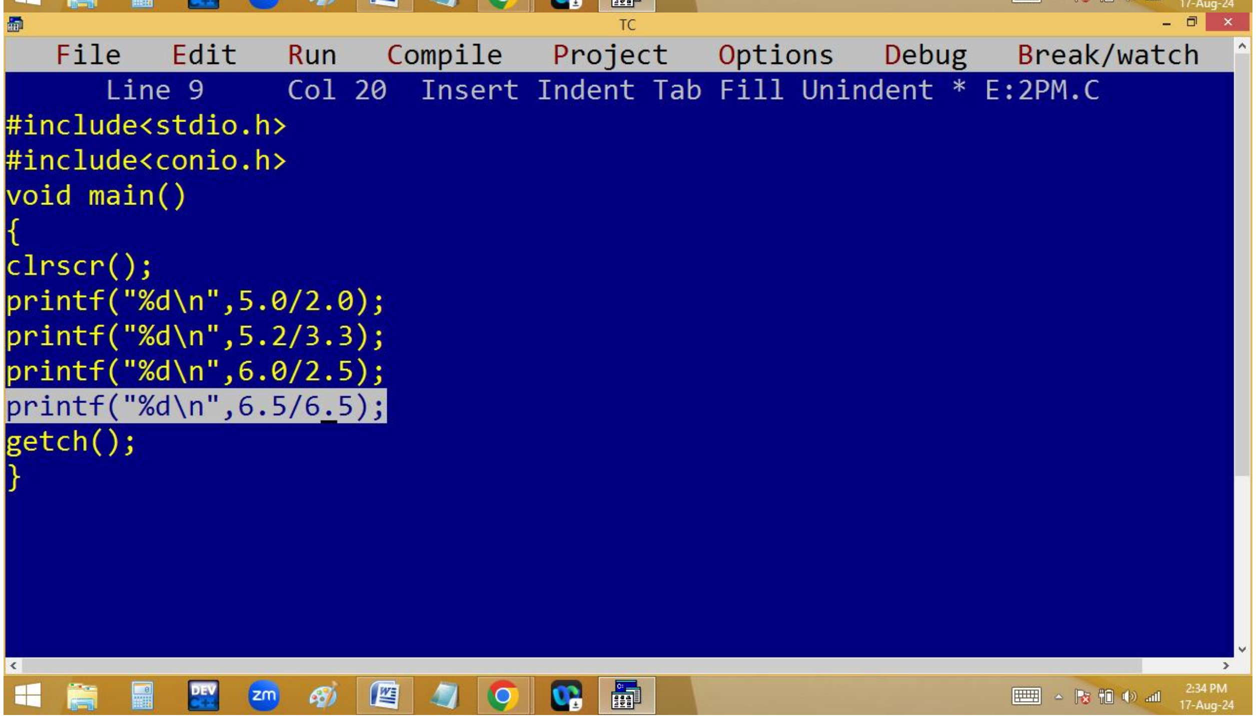
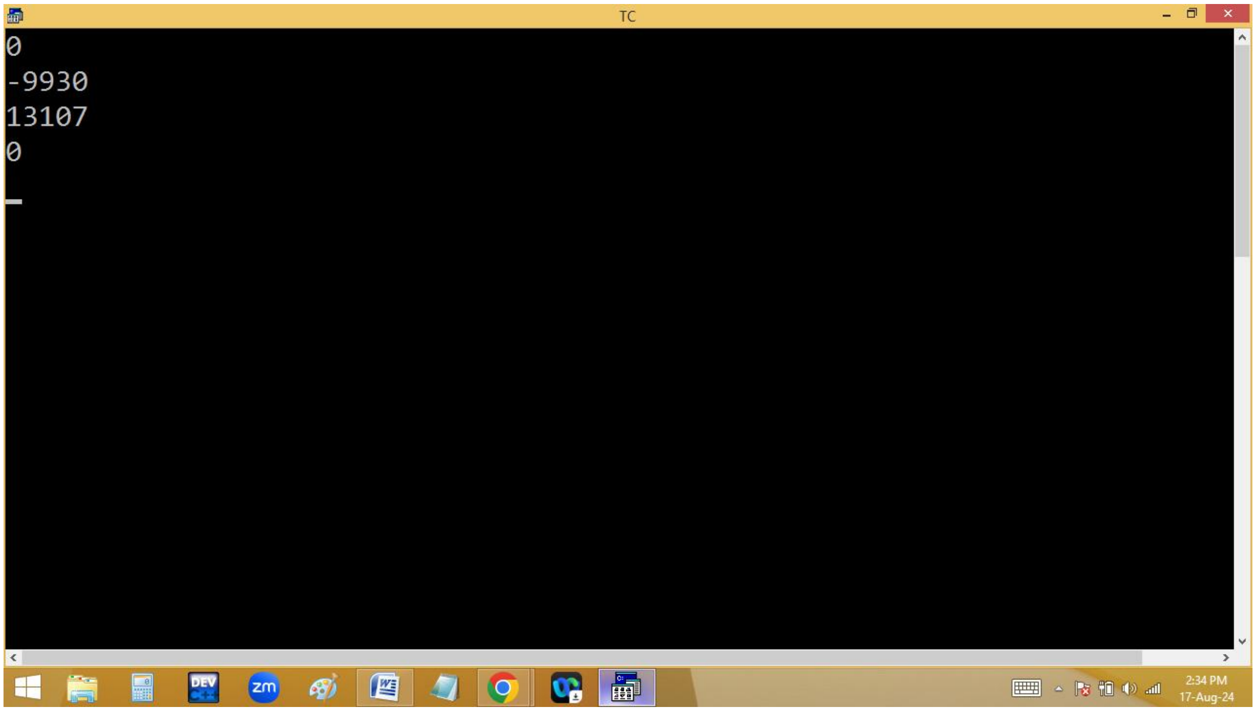


```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 6     Col 40   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5/2); /* runtime error */
}
```



```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 9     Col 20   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5.0/2.0);
printf("%d\n",5.2/3.3);
printf("%d\n",6.0/2.5);
printf("%d\n",6.5/6.5);
getch();
}
```


TC
0
-9930
13107
0
2:34 PM
17-Aug-24

TC

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Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5.0/2.0);
printf("%d\n",5.2/3.3);
printf("%d\n",6.0/2.5);
printf("%d\n",6.5/6.5);
printf("%d\n",6.5);
printf("%d\n",6.0);
printf("%d\n",6.6);
getch();
}
```

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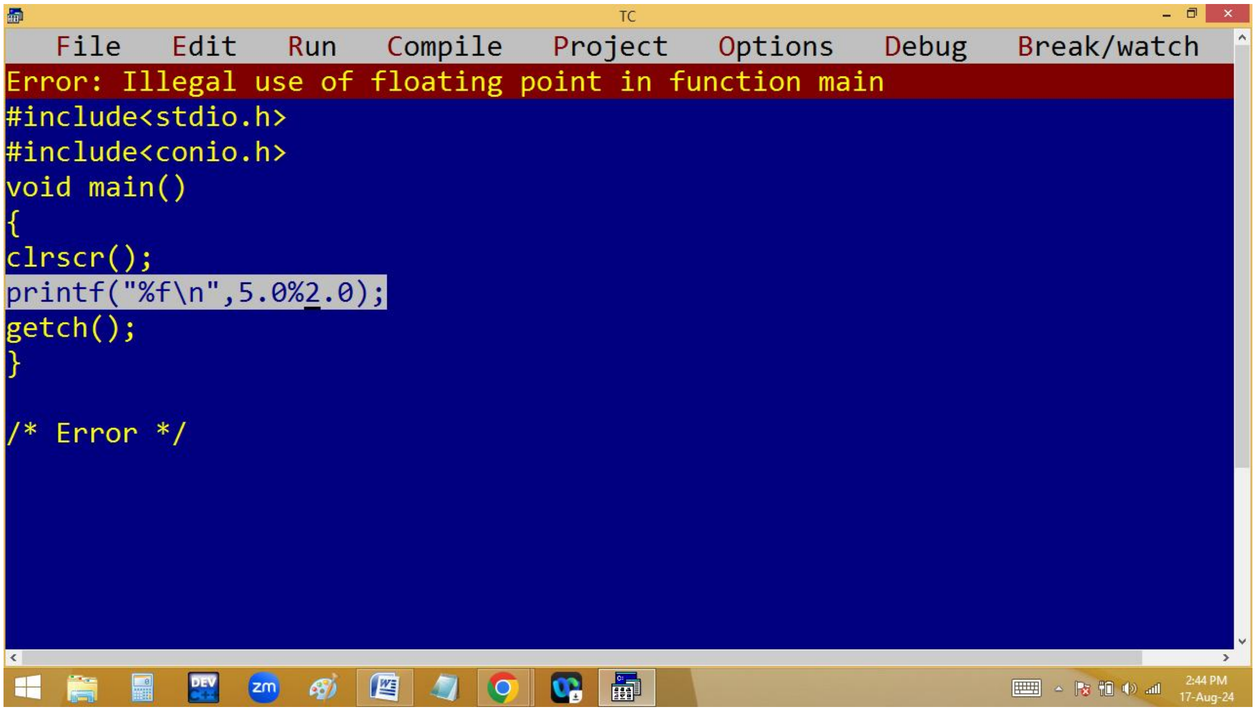
TC

```
0
-9930
13107
0
0
0
26214
```

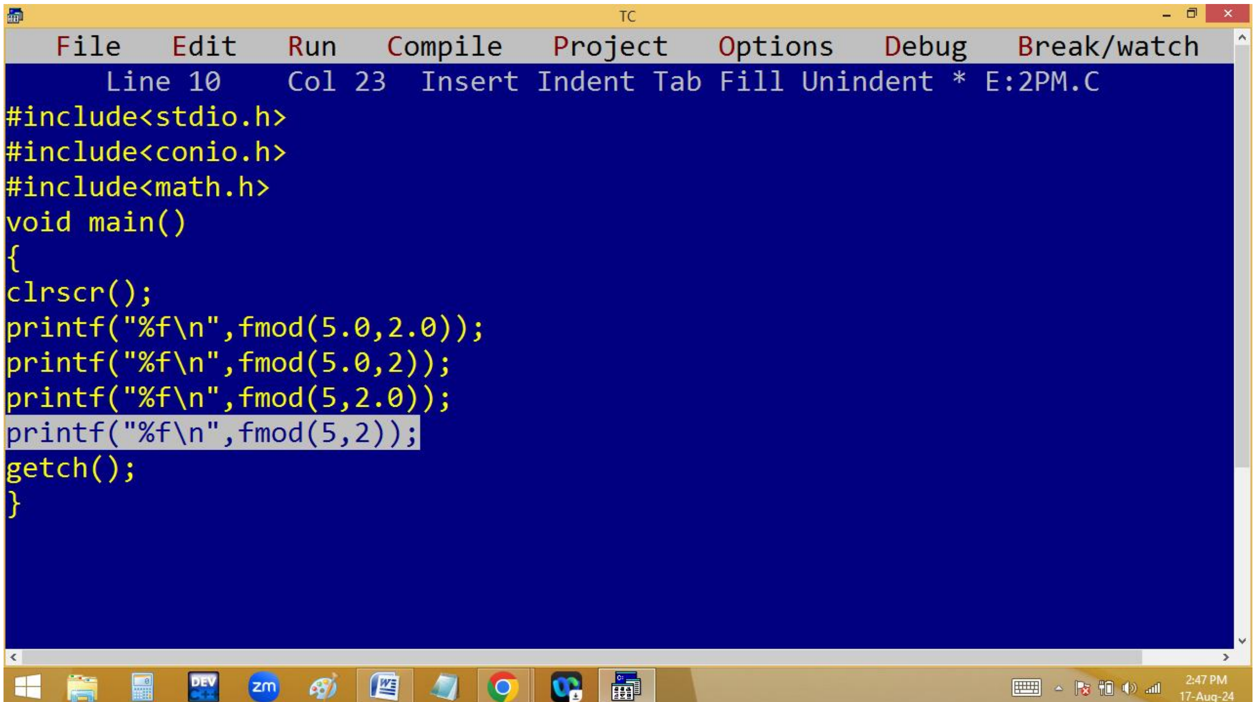
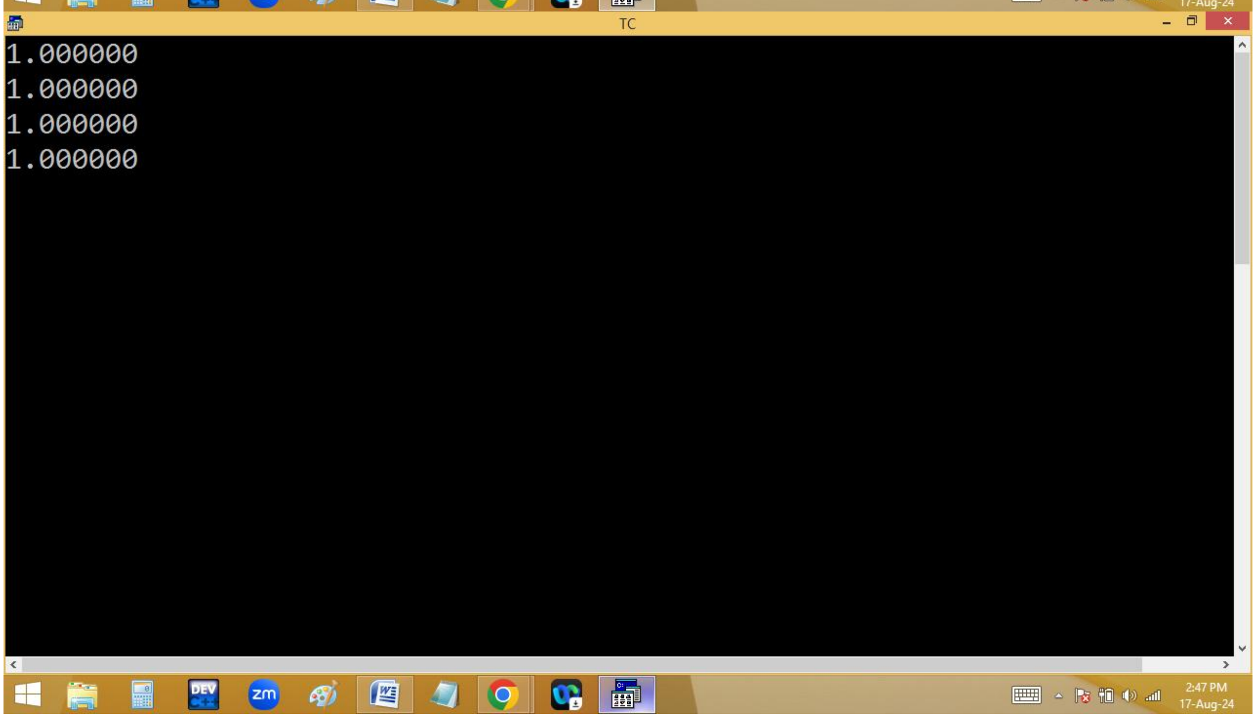
2:36 PM
17-Aug-24

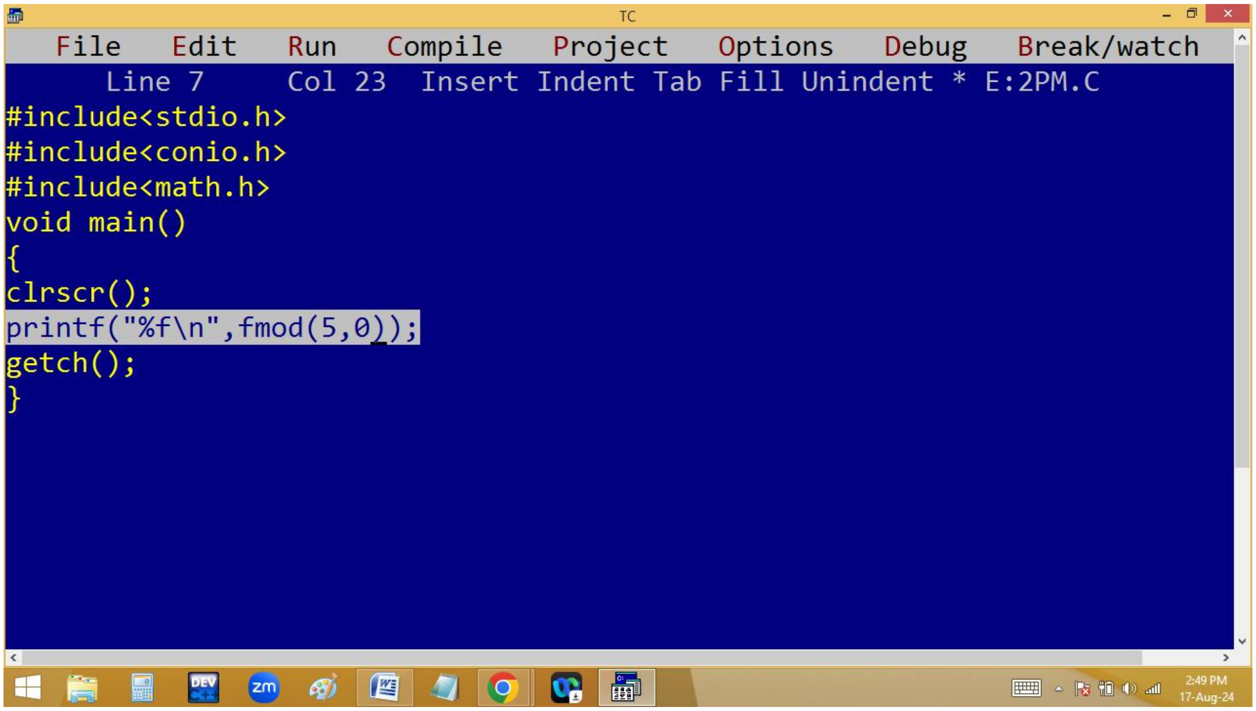
```
   File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 11    Col 11  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float    a = 3; /* impicit type casting */
int      b = 1.2; /* implicit type casting */
clrscr();
printf("%f\n",(float)5/2); /* explicit type casting */
printf("%f\n",5/(float)2); /* explicit type casting */
printf("%f\n",(float)(5/2)); /* explicit type casting */
printf("%d\n",(int)5.0/2); /* explicit type casting */
printf("%f\n",a);
printf("%f\n",a);
getch();
}
```

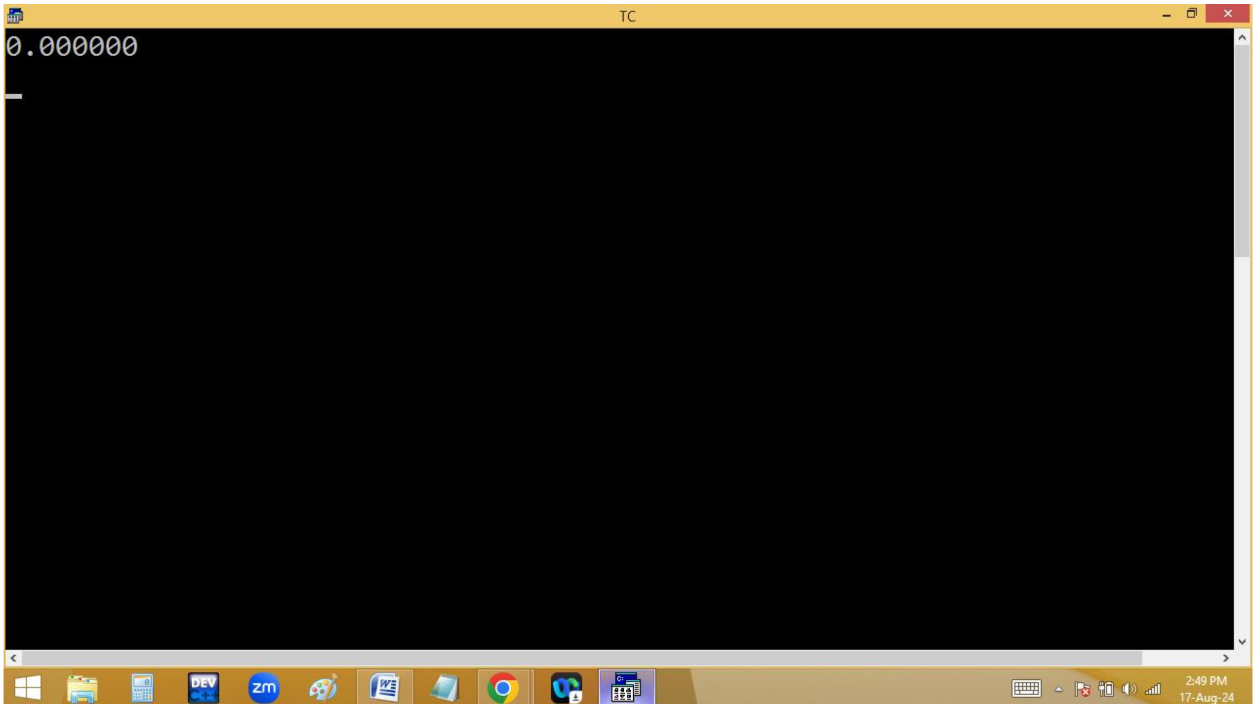
```
2.500000
2.500000
2.000000
2
3.000000
3.000000
```

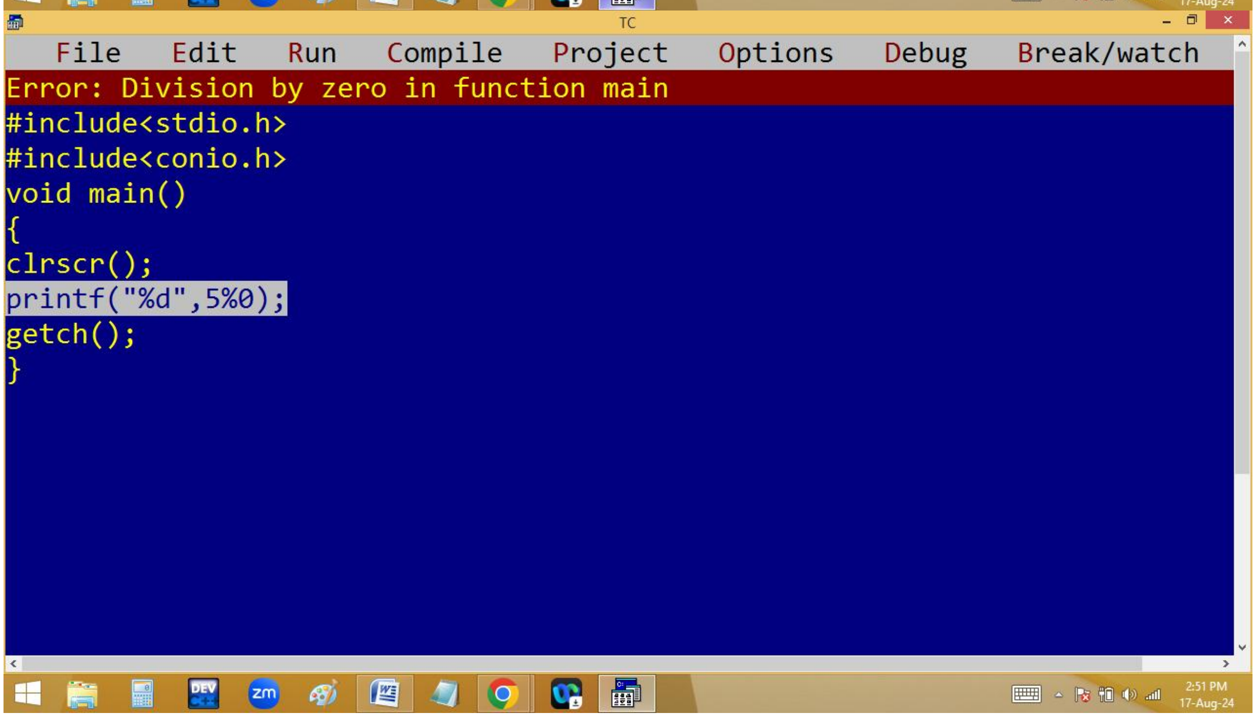


Note: in c & c++ we can't do floating modules with % operator. For this we have to use fmod() available in <math.h>


TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5.0,2.0));
printf("%f\n",fmod(5.0,2));
printf("%f\n",fmod(5,2.0));
printf("%f\n",fmod(5,2));
getch();
}
2:47 PM
17-Aug-24

TC
1.000000
1.000000
1.000000
1.000000
2:47 PM
17-Aug-24


TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5,0));
getch();
}
2:49 PM
17-Aug-24





123/10=12

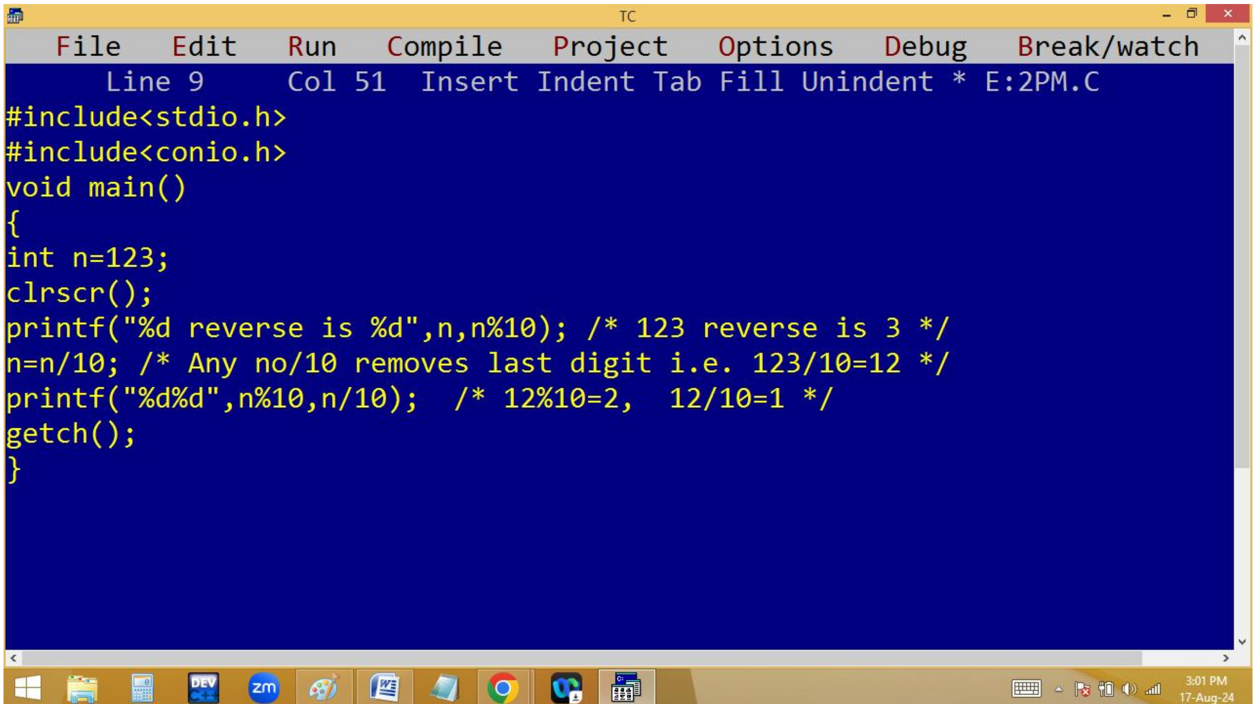
12/10=1

1/10=0

Note: Any no/10 removes last digit.

Write a c program to print a 3 digit no in reverse order without using loop:

Eg: 123 reverse is 321


TC
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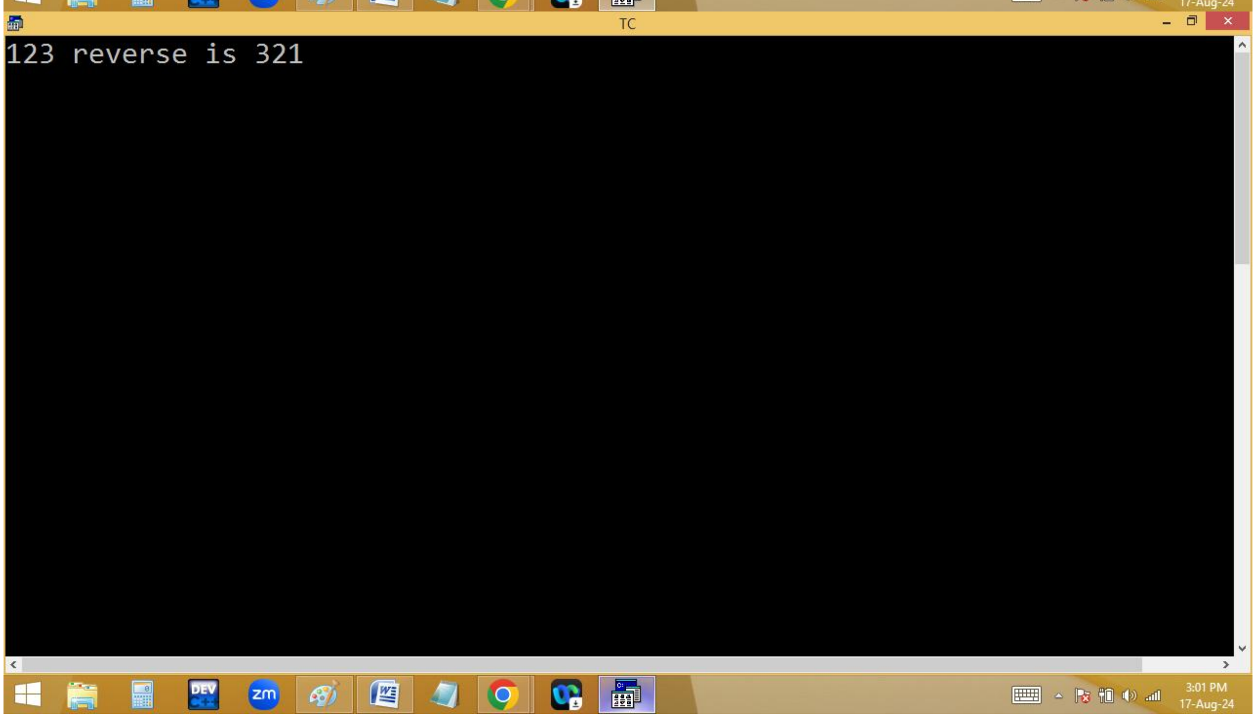
Line 9 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n=123;
clrscr();
printf("%d reverse is %d",n,n%10); /* 123 reverse is 3 */
n=n/10; /* Any no/10 removes last digit i.e. 123/10=12 */
printf("%d%d",n%10,n/10); /* 12%10=2, 12/10=1 */
getch();
}
3:01 PM
17-Aug-24

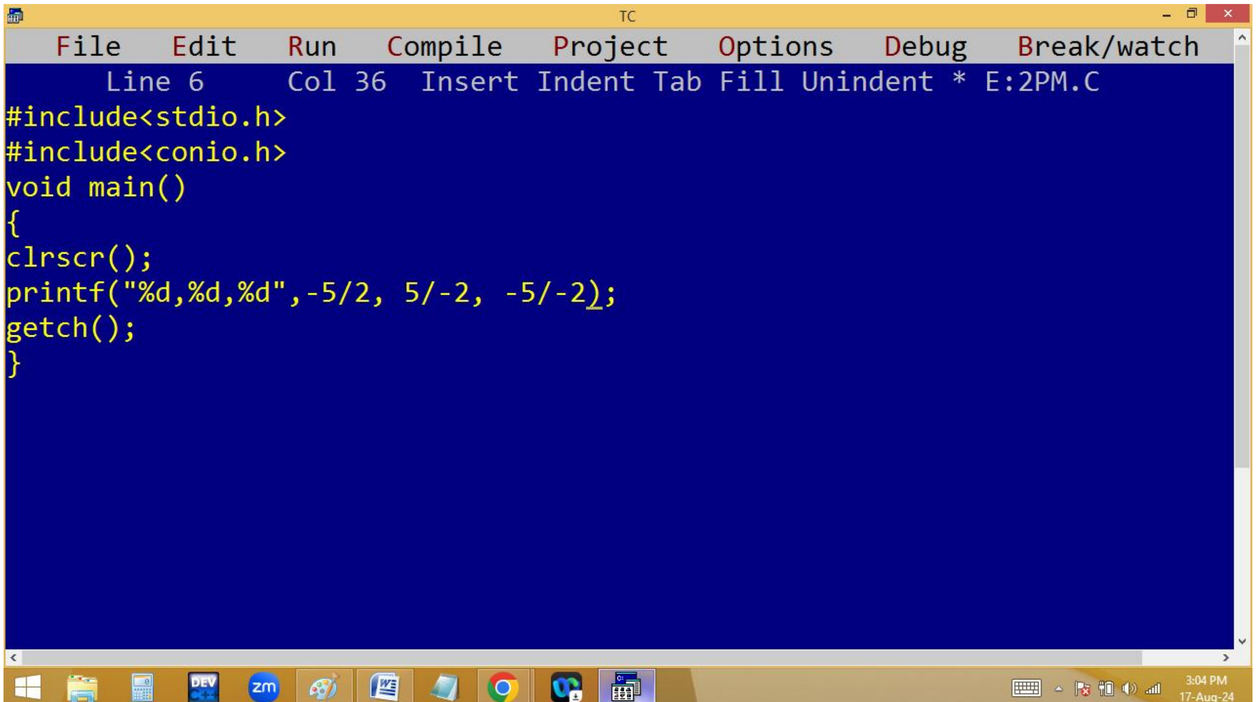

TC
123 reverse is 321
3:01 PM
17-Aug-24

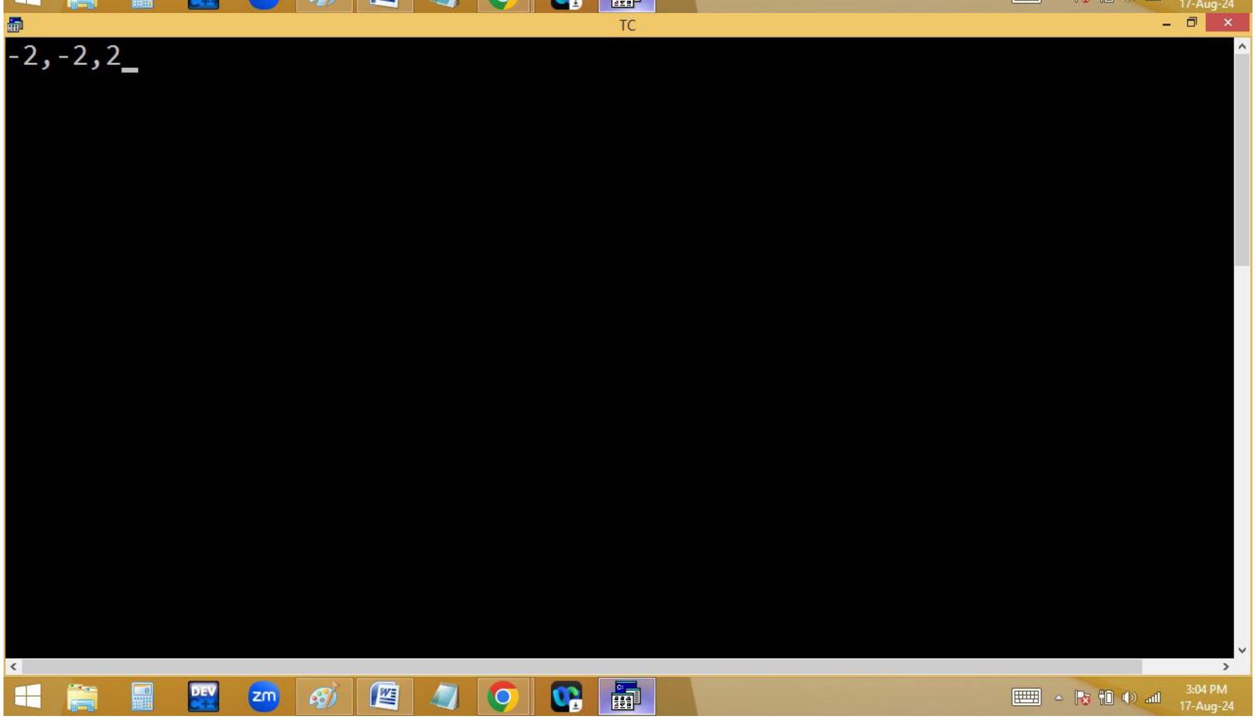
-5/2= -2

5/-2= -2

-5/-2= 2

Note: In division any one operand is negative then result also negative. If both are negative result is positive.





Relational operators [**== (comparison), <, >, <=, >=, != (not equal)**]:

They are used to check the given condition / expression is true or false. If condition is true always it return 1 and false it return 0.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5==5);
printf("%d\n",5 == -5);
printf("%d\n",5.5 == 5.5);
printf("%d\n",5 == 5.0);
printf("%d\n",'5' == 5);
printf("%d\n","a" == "a");
printf("%d\n",'a' == 97);
printf("%d\n",'a'-32 == 'A');    /* 65==65 */
printf("%d\n",5==5==5);
printf("%d\n",5==5==1);
getch();
}
```

```
1
0
1
1
0
0
1
1
0
1
```

```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 10     Col 21   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",'a'>'A');  /* 97 > 65 */
printf("%d\n",'b'<='b');
printf("%d\n",'0'>=0); /* 48 >= 0 */
printf("%d\n",9 != 9);
getch();
}
```

```
1
1
1
0
```

Operator precedence / Operator priority

(ASSOCIATION OF OPERATORS)

1. ()
2. +, - , ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, / , %
5. +, - (Binary)
6. = =, !=
7. &&
8. | |
9. **? :** (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)

TC

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Line 14 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5+3/2==4);
printf("%d\n", (5+3)/2==4);
printf("%d\n", 5-3+2==4);
printf("%d\n", 5%3/2==1);
printf("%d\n", 2+3*4+5==19);
printf("%d\n", 2+3*4+5==25);
printf("%d\n", 2+3*4+5==45);
printf("%d\n", (2+3)*(4+5)==45);
getch();
}
```

TC

```
0
1
1
1
1
0
0
1
```